SYSTEM AND METHOD FOR SECURE COMMUNICATION VIA DISPOSABLE ENCODING MAPS

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Title: SYSTEM AND METHOD FOR SECURE COMMUNICATION VIA DISPOSABLE ENCODING MAPS

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Abstract

A system and method for establishing a secure communication channel by using a temporary, session-specific, randomized data encoding map, effectively creating a form of physical layer encryption. The method comprises: securely sharing a cryptographic seed between two communication endpoints via a secure out-of-band channel; each endpoint independently generating an identical, pseudo-random encoding map based on the shared seed; using the generated map for data transmission over a high-cardinality parallel interface for a single communication session; and destroying the map upon session termination. Because the map linking data to physical signals is known only to the endpoints and exists only for the duration of the session, an eavesdropper intercepting the physical signals cannot decode the data without the map, rendering traditional decryption unnecessary for confidentiality.

Background of the Invention

Securing data in transit has traditionally relied on cryptographic algorithms applied at the upper layers of the OSI model (e.g., TLS/SSL at the Transport/Session layers). This process involves encrypting the plaintext data into ciphertext, transmitting the ciphertext, and decrypting it at the destination. This paradigm assumes that the underlying physical layer is an insecure, open channel. However, the development of high-cardinality communication channels, where the physical signals themselves can be varied in millions of ways, opens the door to a new security model where the encoding of data into physical signals can itself be made secret.

Summary of the Invention

The present invention is a system and method for secure communication that moves the security function to the physical layer. Instead of encrypting the data and sending it over a public channel, this method makes the channel's "language" itself a secret. This is achieved through the use of disposable, session-specific encoding maps.

The process begins with two communicating parties securely sharing a secret seed value through an out-of-band channel (e.g., a pre-shared key, or a key exchange protocol). Both parties then use a deterministic algorithm, such as a seeded pseudo-random number generator, to independently generate an identical, large, and complex encoding map. This map creates a random but agreed-upon correspondence between data blocks and the physical states of the communication channel (e.g., mapping 3-byte data blocks to random 24-bit pixel colors).

The parties then use this unique, temporary map for their communication session. An eavesdropper on the physical link would only capture a stream of seemingly random pixel colors or physical states. Without access to the secret map, the captured signals are indecipherable. Upon session termination, both parties discard the map. A new map is generated for each new session, ensuring that compromising one session does not compromise any others.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the lifecycle of a disposable encoding map for a secure communication session.

Detailed Description of the Invention

Referring to **FIG. 1**, the lifecycle of the secure communication method is shown. The process begins with two endpoints securely sharing a secret seed (101) via an out-of-band channel. Each endpoint then independently uses the seed to computationally generate an identical, pseudo-random, session-specific encoding map (102). A communication session is then initiated (103) using this disposable map to encode and decode all transmitted data. During the session, an eavesdropper (104) may intercept the physical signals, but without the map, the data is indecipherable. The session continues until it is terminated (105). Upon termination, both endpoints securely delete and destroy the disposable encoding map (106), ensuring the secret "language" of that session can never be used again.

Claims

What is claimed is:

1. A method for secure communication between a first device and a second device over a parallel communication channel, the method comprising the steps of:

    - generating a session-specific data encoding map based on a shared secret known to the first and second devices;

- transmitting data from the first device to the second device, wherein the data is encoded into a sequence of physical channel states according to the session-specific map;
- receiving the physical channel states at the second device and decoding the data according to the session-specific map; and
- deleting the session-specific map on at least one of the devices upon termination of a communication session.

2. The method of claim 1, wherein the shared secret is a cryptographic seed, and the map is generated using a deterministic pseudo-random algorithm.

3. The method of claim 1, wherein the shared secret is exchanged via a secure out-of-band channel.

4. The method of claim 1, wherein the parallel communication channel has a cardinality greater than 2.

5. The method of claim 4, wherein the channel comprises a plurality of pixels, and the map creates a correspondence between multi-byte data blocks and unique pixel colors.

6. The method of claim 1, wherein an eavesdropper intercepting the sequence of physical channel states without the session-specific map is unable to decode the data.

7. The method of claim 1, wherein the parallel communication channel is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.

Drawings

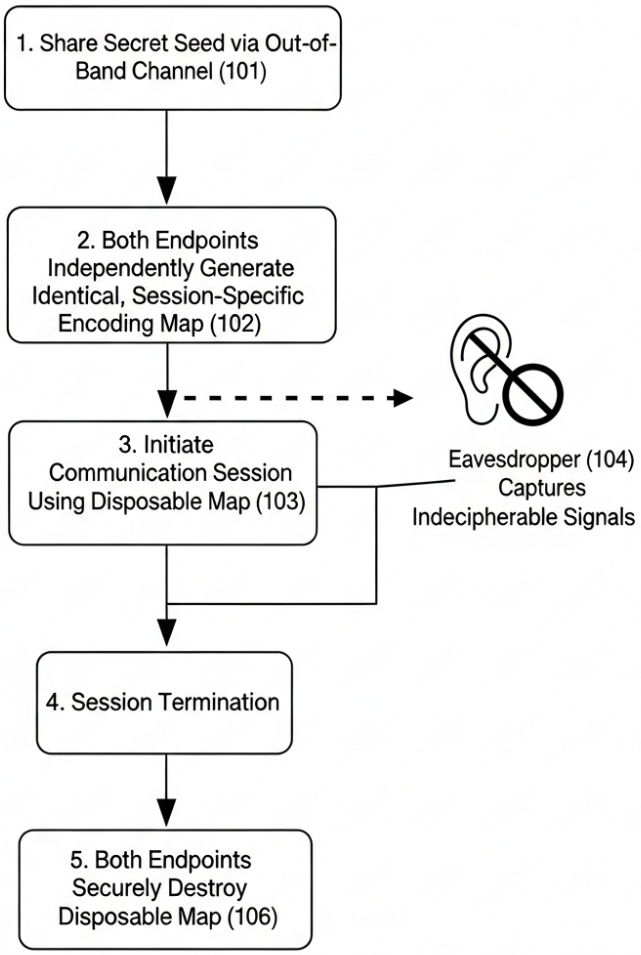


Figure 1: Flowchart illustrating the lifecycle of a disposable encoding map for a secure communication session.